

# Class 10: Structural Bioinformatics 1

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## PDB Statistics

The Protein Data Bank (PDB) is the main repository of biomolecular structures. Let's see what it contains:

```
stats <- read.csv("Data Export Summary.csv")
stats
```

|   | Molecular.Type          | X.ray   | EM      | NMR    | Integrative | Multiple.methods |
|---|-------------------------|---------|---------|--------|-------------|------------------|
| 1 | Protein (only)          | 178,795 | 21,825  | 12,773 | 343         | 226              |
| 2 | Protein/Oligosaccharide | 10,363  | 3,564   | 34     | 8           | 11               |
| 3 | Protein/NA              | 9,106   | 6,335   | 287    | 24          | 7                |
| 4 | Nucleic acid (only)     | 3,132   | 221     | 1,566  | 3           | 15               |
| 5 | Other                   | 175     | 25      | 33     | 4           | 0                |
| 6 | Oligosaccharide (only)  | 11      | 0       | 6      | 0           | 1                |
|   | Neutron                 | Other   | Total   |        |             |                  |
| 1 | 84                      | 32      | 214,078 |        |             |                  |
| 2 | 1                       | 0       | 13,981  |        |             |                  |
| 3 | 0                       | 0       | 15,759  |        |             |                  |
| 4 | 3                       | 1       | 4,941   |        |             |                  |
| 5 | 0                       | 0       | 237     |        |             |                  |
| 6 | 0                       | 4       | 22      |        |             |                  |

```
stats$X.ray
```

```
[1] "178,795" "10,363" "9,106" "3,132" "175" "11"
```

```
sum(stats$Neutron)
```

```
[1] 88
```

The comma in these numbers leads to the numbers here being read as characters.

```
c(100, 10, "barry")
```

```
[1] "100"  "10"   "barry"
```

```
library(readr)
stats <- read_csv("Data Export Summary.csv")
```

Rows: 6 Columns: 9

-- Column specification -----

Delimiter: ","

chr (1): Molecular Type

dbl (4): Integrative, Multiple methods, Neutron, Other

num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
stats
```

# A tibble: 6 x 9

|   | `Molecular Type`<br><chr> | `X-ray`<br><dbl> | EM<br><dbl> | NMR<br><dbl> | Integrative<br><dbl> | `Multiple methods`<br><dbl> | Neutron<br><dbl> |
|---|---------------------------|------------------|-------------|--------------|----------------------|-----------------------------|------------------|
| 1 | Protein (only)            | 178795           | 21825       | 12773        | 343                  | 226                         | 84               |
| 2 | Protein/Oligosacch~       | 10363            | 3564        | 34           | 8                    | 11                          | 1                |
| 3 | Protein/NA                | 9106             | 6335        | 287          | 24                   | 7                           | 0                |
| 4 | Nucleic acid (only)       | 3132             | 221         | 1566         | 3                    | 15                          | 3                |
| 5 | Other                     | 175              | 25          | 33           | 4                    | 0                           | 0                |
| 6 | Oligosaccharide (o~       | 11               | 0           | 6            | 0                    | 1                           | 0                |

# i 2 more variables: Other <dbl>, Total <dbl>

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy? 93.79%

```
sum(c(stats$`X-ray`, stats$`EM`))
```

```
[1] 233552
```

```
sum(stats$Total)
```

```
[1] 249018
```

```
(sum(c(stats$`X-ray`, stats$`EM`)))/sum(stats$Total)
```

```
[1] 0.937892
```

```
n.xray <- sum(stats$`X-ray`)
n.em <- sum(stats$`EM`)
n.total <- sum(stats$Total)
(n.xray + n.em)/n.total
```

```
[1] 0.937892
```

Q2: What proportion of structures in the PDB are protein? 85.97%

```
n.protein <- stats[1, "Total"]
p_protein <- n.protein/n.total
p_protein
```

```
      Total
1 0.8596889
```

Q3. Skip...

## Visualizing the HIV-1 protease structure

We can use the Molstar viewer online: <http://molstar.org/viewer/>

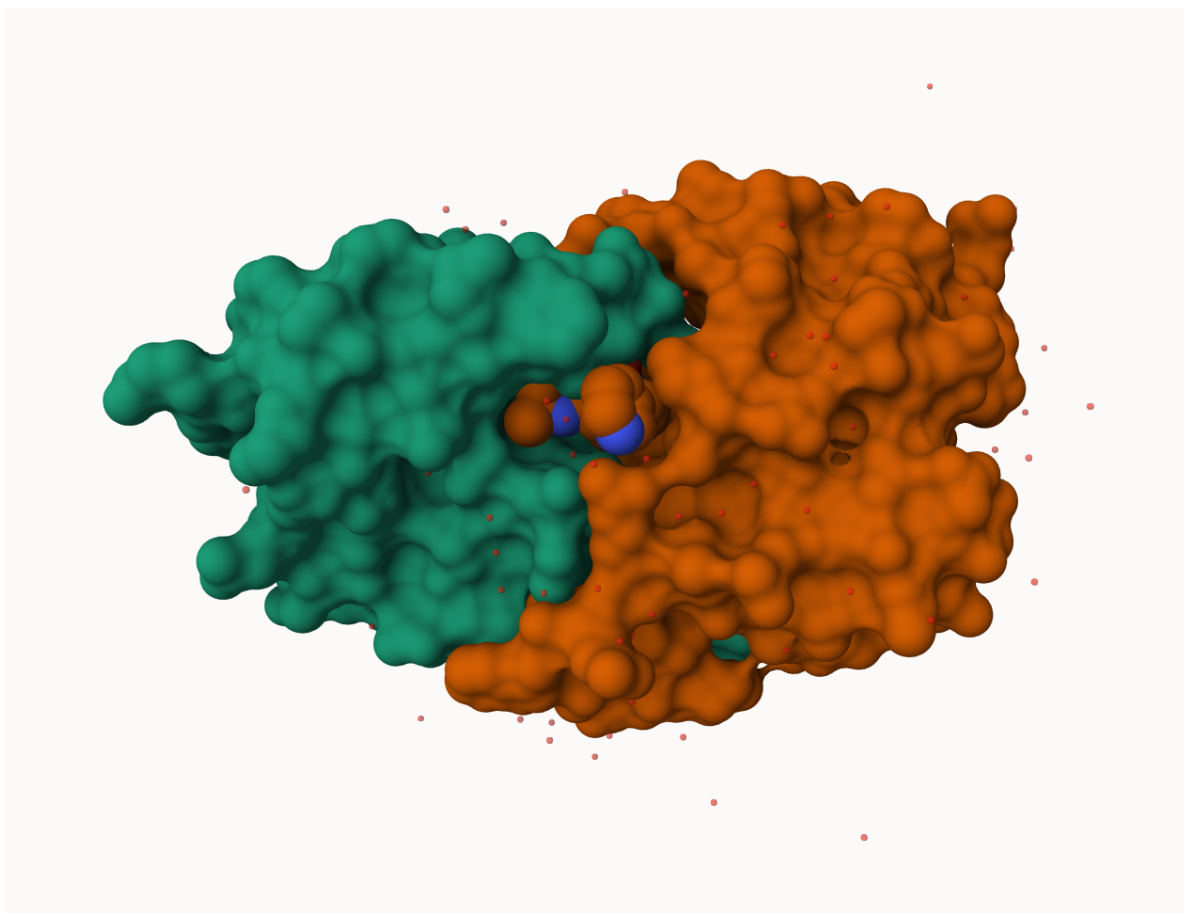
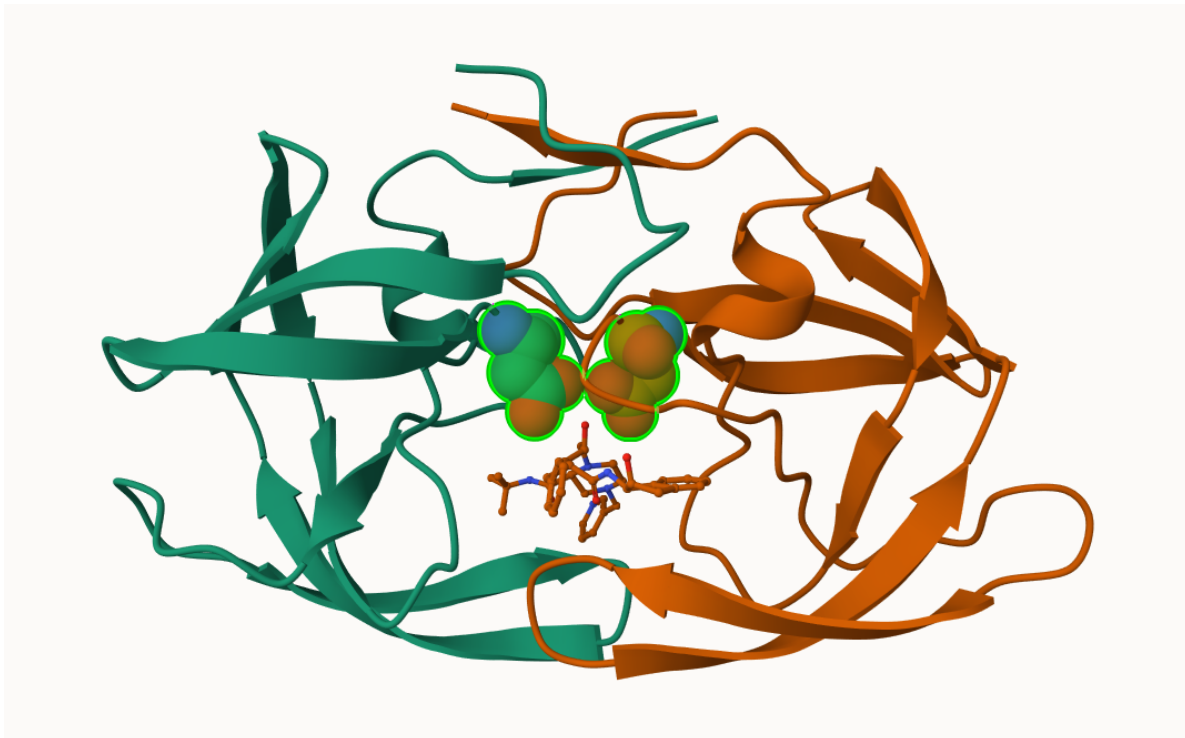


Figure 1: My first image of HIV-Pr with surface display showing ligand

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure? X-ray crystallography detects electron density, not atom directly. Oxygen has many electrons while hydrogen atoms have only one electron.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?  
HOH 308

A new clean image showing the catalytic ASP25 amino acids in both chains of the HIV-PR dimer along with the inhibitor and all important active site water.



Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

### Bio3D package for structural bioinformatics

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```

Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

```

Protein sequence:

```

PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF

```

```

+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call

```

```
head(pdb$atom)
```

|   | type | eleno | elety | alt  | resid | chain | resno | insert | x      | y      | z     | o | b     |
|---|------|-------|-------|------|-------|-------|-------|--------|--------|--------|-------|---|-------|
| 1 | ATOM | 1     | N     | <NA> | PRO   | A     | 1     | <NA>   | 29.361 | 39.686 | 5.862 | 1 | 38.10 |
| 2 | ATOM | 2     | CA    | <NA> | PRO   | A     | 1     | <NA>   | 30.307 | 38.663 | 5.319 | 1 | 40.62 |
| 3 | ATOM | 3     | C     | <NA> | PRO   | A     | 1     | <NA>   | 29.760 | 38.071 | 4.022 | 1 | 42.64 |
| 4 | ATOM | 4     | O     | <NA> | PRO   | A     | 1     | <NA>   | 28.600 | 38.302 | 3.676 | 1 | 43.40 |
| 5 | ATOM | 5     | CB    | <NA> | PRO   | A     | 1     | <NA>   | 30.508 | 37.541 | 6.342 | 1 | 37.87 |
| 6 | ATOM | 6     | CG    | <NA> | PRO   | A     | 1     | <NA>   | 29.296 | 37.591 | 7.162 | 1 | 38.40 |

|   | segid | elesy | charge |
|---|-------|-------|--------|
| 1 | <NA>  | N     | <NA>   |
| 2 | <NA>  | C     | <NA>   |
| 3 | <NA>  | C     | <NA>   |
| 4 | <NA>  | O     | <NA>   |
| 5 | <NA>  | C     | <NA>   |
| 6 | <NA>  | C     | <NA>   |

Q7: How many amino acid residues are there in this pdb object? 198

Q8: Name one of the two non-protein residues? HOH

Q9: How many protein chains are in this structure? 2

```

library(bio3dview)
library(NGLViewer)

# view.pdb(pdb) |> setSpin()

# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation

# view.pdb(pdb, cols=c("navy","teal"), highlight = sele, highlight.style = "spacefill") |>

```

## Predicting functional motions of a single structure

Read an ADK structure from the PDB database:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file  
 PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

Call: read.pdb(file = "6s36")

```

Total Models#: 1
  Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

```

Protein sequence:

```

MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
TDELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

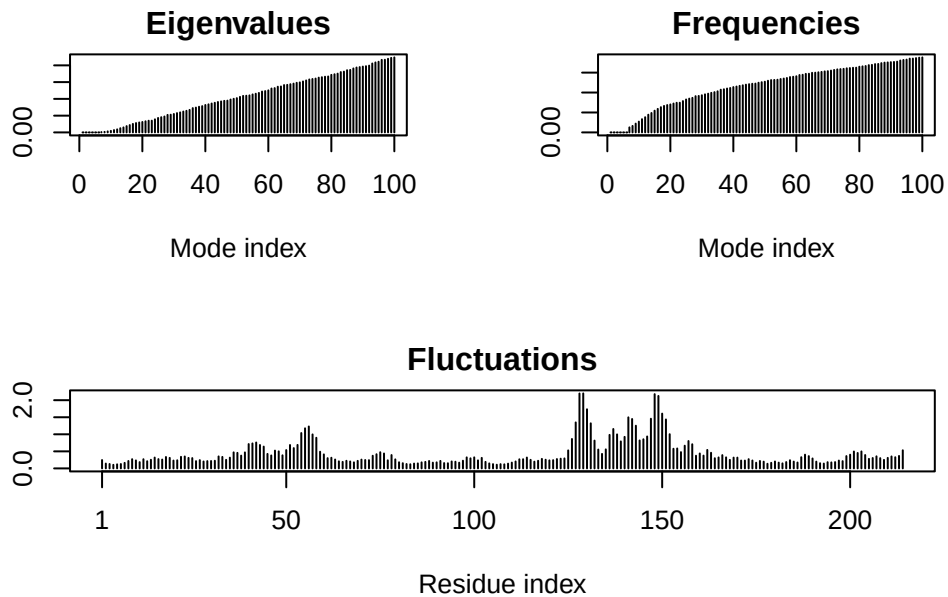
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.014 seconds.  
Diagonalizing Hessian... Done in 0.054 seconds.
```

```
plot(m)
```



Write out our results as a wee trajectory/movie of predicted motions:

```
# mktrj(m, file="adk_m7.pdb")
```

## Comparative structure analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?  
msa

Q11. Which of the above packages is not found on BioConductor or CRAN?  
bio3dview



Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket? True

```
library(bio3d)
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake\_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120

      121      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTPALIG
      121      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?  
214